

Architecture Deep Dive

This presentation covers our system architecture and implementation details.

Tech Stack

- Backend: Node.js + Express
- Database: PostgreSQL
- Cache: Redis
- Message Queue: RabbitMQ

Data Processing

```
interface DataRecord {  
  id: string;  
  value: number;  
  timestamp: Date;  
}  
  
async function processBatch(records: DataRecord[]): Promise<void> {  
  const validated = records.filter(validate);  
  const transformed = await transformData(validated);  
  await saveToDatabase(transformed);  
}
```

Performance Metrics

Operation	Latency (p50)	Throughput
Read	12ms	5K req/s
Write	18ms	3K req/s
Update	15ms	4K req/s

Database Schema

```
CREATE TABLE users (  
  id UUID PRIMARY KEY,  
  email VARCHAR(255) UNIQUE NOT NULL,  
  created_at TIMESTAMP DEFAULT NOW(),  
  metadata JSONB  
);
```

Monitoring

- Metrics: Prometheus + Grafana
- Logging: ELK Stack
- Tracing: Jaeger
- Cache hit rate: 94.3%

Next Steps

1. Complete API v3 migration
2. Implement GraphQL layer
3. Enhance monitoring dashboards

Questions

GitHub: github.com/company/repo

Docs: docs.company.com